

Causal Inference Crash Course:

Part 10 — Conformal Inference

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Causal Inference Series

1. Foundations
2. Causal Models
3. Inference
4. Best Practices
5. Heterogeneous Treatment Effects
6. Panel Models
7. Regression Discontinuity
8. Surrogate Models
9. Arguable Validation
10. **Conformal Inference** ← **this deck**

Overview

- **So what?** Every model's point prediction is wrong by some amount — conformal inference is how you say honestly by how much. It turns *any* prediction into a **set** with a **finite-sample, distribution-free coverage guarantee**, needing only **exchangeability**.
- The payoff: this same machinery gives **valid causal inference with a single treated unit** (synthetic control), where classical standard errors go quiet.
- We cover:
 - the idea: **split conformal** and its coverage guarantee
 - the one assumption — **exchangeability** — and what breaks it
 - two fixes when it fails: **moving blocks** and **weighting**
 - the causal payoff: conformal inference for **synthetic controls**
- Assumes the earlier **Inference** and **Panel Models** decks.

The Idea

You don't have to trust the model — you just have to believe the data's story doesn't depend on the order you saw it in.

What we want

- Here's the uncomfortable question nobody asks about a point prediction: how often is it *this* far off? Most workflows don't say — conformal inference answers it honestly, without pretending the model is right.
- Classical prediction intervals need the **model to be right** — Wald wants a correct parametric form, Bayes a correct likelihood. Neither survives contact with the black-box model you actually shipped.
- **Conformal** asks for far less: a set $\mathcal{C}(x)$ with

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha$$

in **finite samples**, for **any** distribution, wrapped around **any** predictor — however ugly.

- Running example we'll carry through: predict a customer's next-quarter purchases, and give an interval that really covers 90% of the time.

The one assumption: exchangeability

- Here's the trade: give up on knowing the model is right, and in exchange you only need to believe the data's underlying pattern doesn't care about order. That's a much easier thing to defend.
- Formally: $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ are **exchangeable** — their joint distribution is unchanged by reordering them. (i.i.d. is the familiar special case.)
- That's the *whole* assumption — no distributional form, no correctly specified model. Everything downstream is a consequence of it.
- And it's exactly what **fails** for time series and panels — the moment your data has a "before" and "after," this assumption is in trouble. We'll come back to that.

Any Score Works — a Good One Just Tightens the Set

- Before you can say "how often are we wrong," you need a ruler for "how wrong." That ruler is the **nonconformity score** $s(x, y)$.
- Regression default: the absolute residual of any fitted $\hat{f}$,

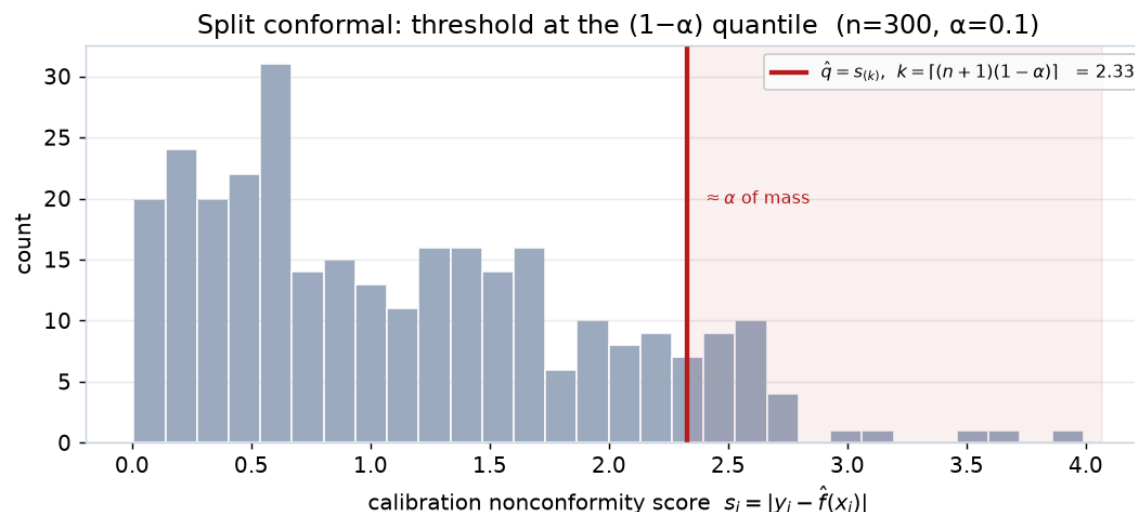
$$s(x, y) = |y - \hat{f}(x)|.$$

- The guarantee holds for **any** score and **any** model — a *good* score doesn't buy validity, you already have that. It only makes the set **tighter**.
- *Details: [Appendix — Choosing a Conformity Score](#)*

Split conformal — the recipe

The mechanics, step by step. For miscoverage level α , split the data into a **training** fold and a **calibration** fold of size n :

1. **Fit** $\hat{f}$ on the training fold only.
2. **Score** the calibration set: $s_i = s(X_i, Y_i)$, $i = 1, \dots, n$.
3. **Quantile**: $\hat{q} = s_{(k)}$, the k -th smallest score, with $k = \lceil (n + 1)(1 - \alpha) \rceil$.
4. **Predict**: $\mathcal{C}(x) = \{y : s(x, y) \leq \hat{q}\}$.



The Guarantee Holds Even When Your Model Is Wrong

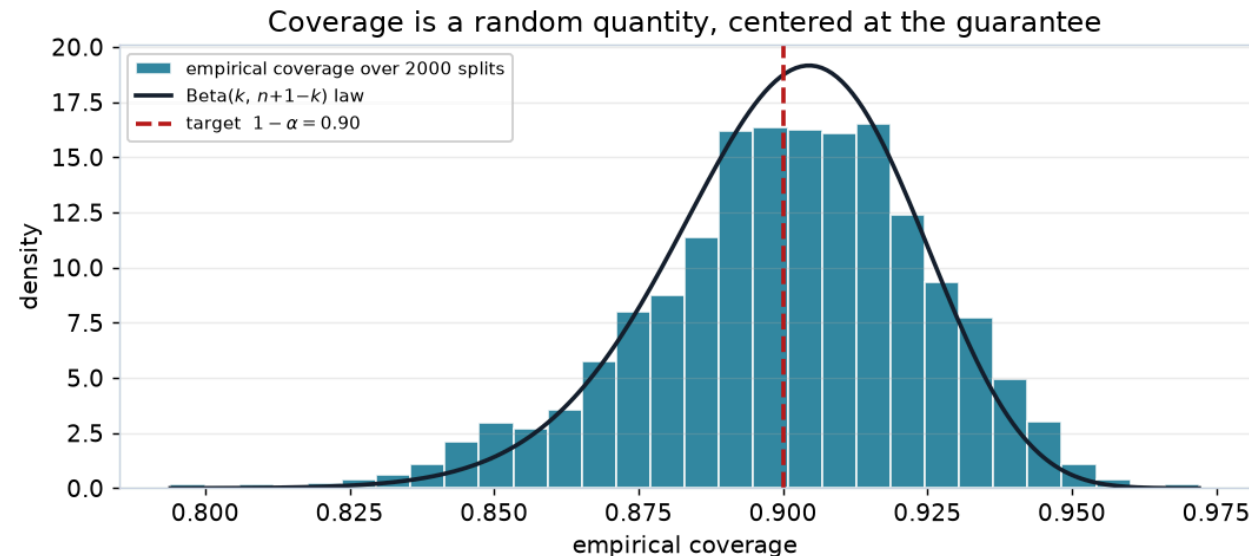
- This is the payoff of the last two slides: a real, provable promise about how often your set is right — not "should be fine in practice."
- **Theorem** (Vovk et al. 2005; Lei et al. 2018). If the data are **exchangeable**, split conformal satisfies

$$1 - \alpha \leq \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \leq 1 - \alpha + \frac{1}{n+1}.$$

- **Finite-sample, distribution-free, model-free.** The upper bound needs only that scores have no ties.
- The model can be **arbitrarily wrong** — coverage still holds. A bad model just gets you an honestly **wide** set, not a broken guarantee.
- Details: [Appendix — Why Coverage Survives a Bad Model](#)

Why it works

- Insert the unseen test score as an $(n+1)$ -th point. By exchangeability its **rank is uniform** on $\{1, \dots, n+1\}$, so it exceeds the k -th calibration value only about α of the time — exactly the miscoverage event.
- So coverage is a **property of the procedure**, not luck: over repeated draws it follows a **Beta($k, n+1-k$)** law centered at $1 - \alpha$.



Conformal Prediction Is Hypothesis Testing in Disguise

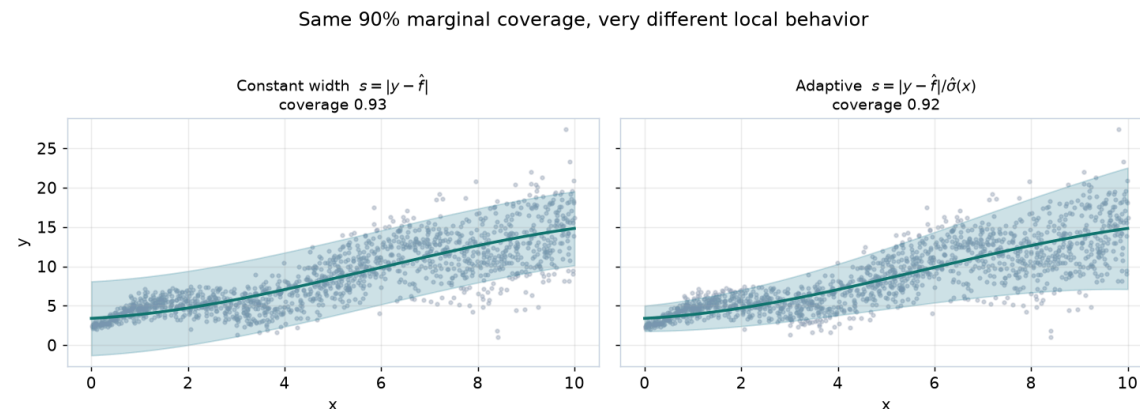
- Equivalent framing (Shafer & Vovk 2008): conformal is **hypothesis testing turned into prediction**. For each candidate label y ,

$$p(y) = \frac{1 + \#\{i : s_i \geq s(x, y)\}}{n + 1}, \quad \mathcal{C}(x) = \{y : p(y) > \alpha\}.$$

- Keep the labels you **cannot reject** at level α .
- This is the view we'll lean on for synthetic control — where the "label" becomes a hypothesized treatment effect, and rejecting it is the whole game.
- *Details: [Appendix — Conformal Sets as Test Inversion](#)*

Adaptivity — and its limit

- **Heads up:** the guarantee we just proved is **marginal**, not conditional — some regions can be **under-covered** and others **over-covered**; exact conditional coverage is impossible distribution-free.
- Fix with a smarter *score*, not a new guarantee: normalize by a local scale $s = |y - \hat{f}(x)| / \hat{\sigma}(x)$, or use **conformalized quantile regression** (Romano, Patterson & Candès 2019).

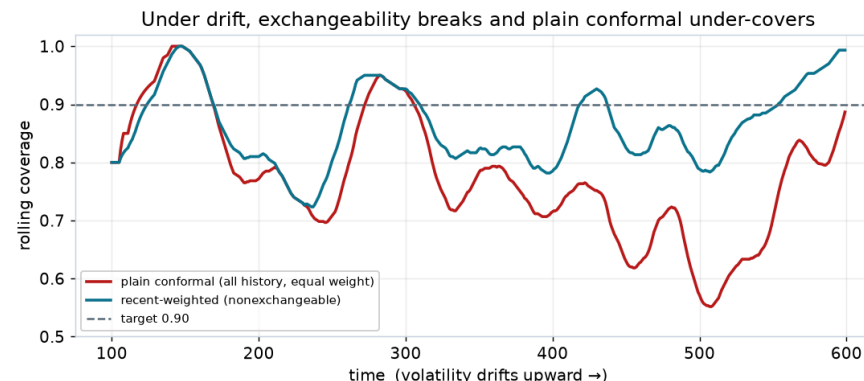


When Exchangeability Fails

Real data has a "before" and "after." That breaks the one assumption we just leaned on — here's what to do about it.

Time and treatment break exchangeability

- Everything so far leaned on one belief: that reordering the data wouldn't change its story. Time series don't grant that — there's a real "before" and "after," and most data we actually work with has one.
- Panels and time series are **serially dependent** and often **drift** — reorder them and the joint law changes, so exchangeability fails.
- Worse for causal work: the null holds only **before** treatment, and the post-period is exactly what we suspect differs.
- Plain conformal then **under-covers** — calibrated on a stale past that no longer resembles the present.



Fix A — permute blocks, not points

- If shuffling the data breaks its story, don't shuffle points — shuffle in chunks big enough to keep the story intact.
- Keep **contiguous blocks** of residuals together and permute the block **positions** — the moving-block idea borrowed from the block bootstrap.
- Local dependence rides along *inside* each block; only its placement is randomized, so the null datasets keep the real time structure.
- **Exact** under exchangeability, **approximately valid** under stationarity and weak dependence — the block length must exceed the dependence range (Chernozhukov, Wüthrich & Zhu 2018, 2021).

Fix B — weight what belongs

- The alternative move: don't chunk the data, just admit some points matter more than others — and say so explicitly, with weights fixed in advance.
- **Nonexchangeable conformal** (Barber, Candès, Ramdas & Tibshirani 2023): attach fixed weights w_i and take **weighted quantiles** of the scores — for time series, weight **recent** points more.

- The guarantee degrades gracefully rather than breaking outright:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha - \Delta,$$

where Δ is a weighted **distance from exchangeability**; it vanishes when the up-weighted points really do resemble the target.

- One rule: the weights must be **fixed a priori** — not learned from the scores, or you're back to cheating.

Two relaxations of one idea

- **Back to the big idea:** both fixes are the same move — restore just enough symmetry for the uniform-rank argument to go through, without pretending the data is something it isn't.
- **Blocks** — the *discrete* relaxation: glue dependent points together so a permutation can't shatter the local structure.
- **Weights** — the *continuous* relaxation: down-weight the points that don't belong with the target.

Conformal Inference for Synthetic Controls

The payoff: turn this same machinery into a causal test that works with just one treated unit.

Causal Inference = Prediction + a Break Test

- Here's the payoff we've been building to: everything up to now was about intervals for predictions. This section turns the same machinery into a **causal test** — with just one treated unit, no fancy standard errors required.
- **Chernozhukov, Wüthrich & Zhu (2021)** glue two familiar problems together: causal inference = **counterfactual prediction** + **a test for a structural break**.
- Predict the treated unit's *untreated* path (synthetic control, DiD, factor models...), then ask whether the **post-treatment residuals break** from the pre-treatment ones.
- The pitch: **inference that doesn't care how good your synthetic control is** — validity comes from the permutation, not from a correct model.
- Details: [Appendix — The CWZ Framework](#)

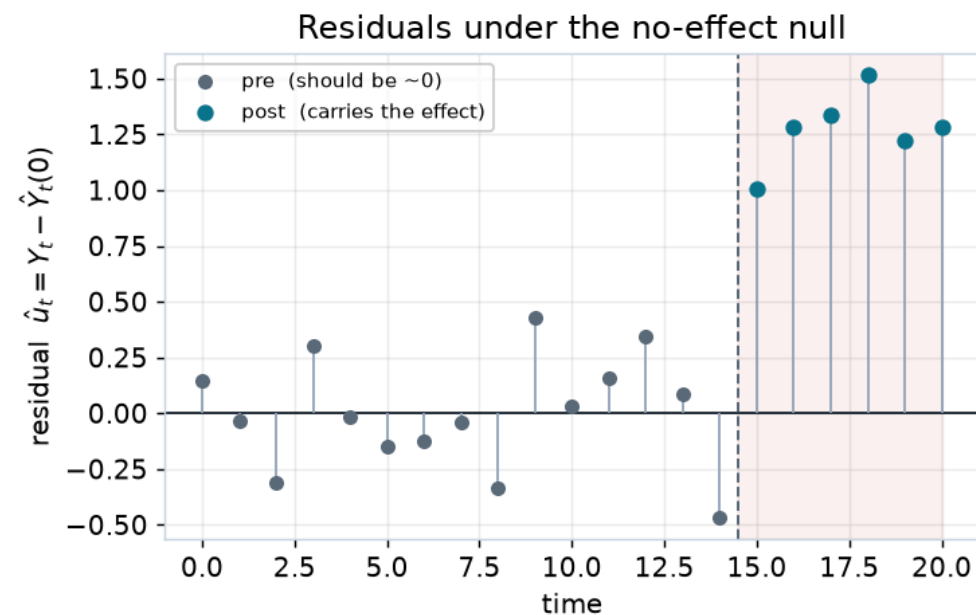
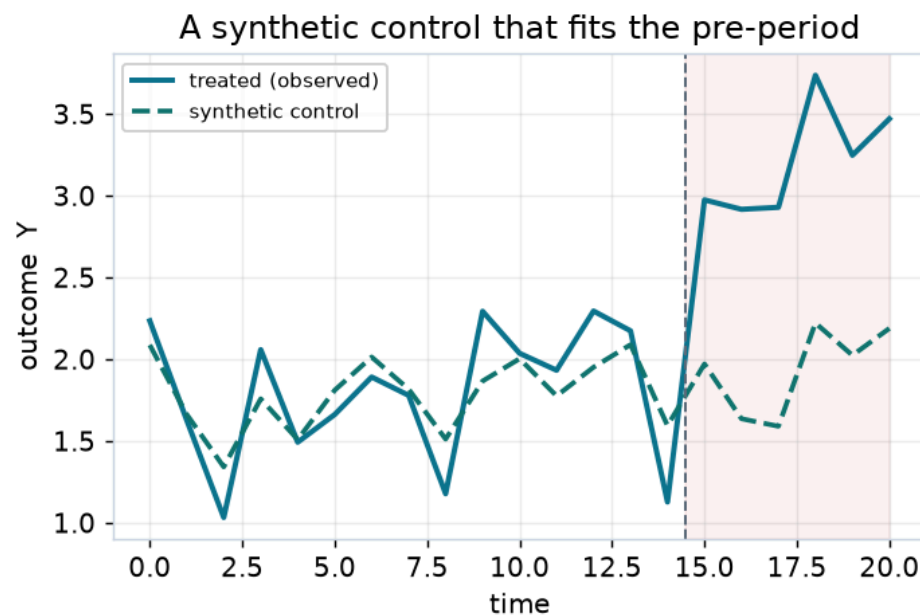
Setup — we predict the counterfactual

- Before the test, we need the ingredients: a treated unit, a donor pool, and a prediction of what would've happened without treatment — nothing new if you sat through the Panel Models deck.
- One treated unit Y_{1t} , a donor pool Y_{0t} , pre-periods $t \leq T_0$ and post-periods $t > T_0$. Fit a **synthetic control** $\hat{Y}_{1t}(0)$ (from the Panel Models deck).
- The **gap** $\hat{u}_t = Y_{1t} - \hat{Y}_{1t}(0)$ is the residual. Under *no effect*, that residual process has **no break** at T_0 .
- The residuals over time are serially dependent — so this is exactly the non-exchangeable case, handled by **moving-block** permutations from the last section.

Step 1 — impose the null, read the residuals

- We're testing a hypothesized effect size θ — a sharp null on the post-period effect, $H_0 : \tau_t = \theta$. Subtract it from the **post outcomes only**, refit over the **whole timeline**, and take residuals for every t :

$$\tilde{Y}_{1t} = Y_{1t} - \theta \cdot \mathbf{1}\{t > T_0\}, \quad \hat{u}_t = \tilde{Y}_{1t} - \hat{Y}_{1t}(0)$$



One Number, Tested Against Its Shuffled Twins

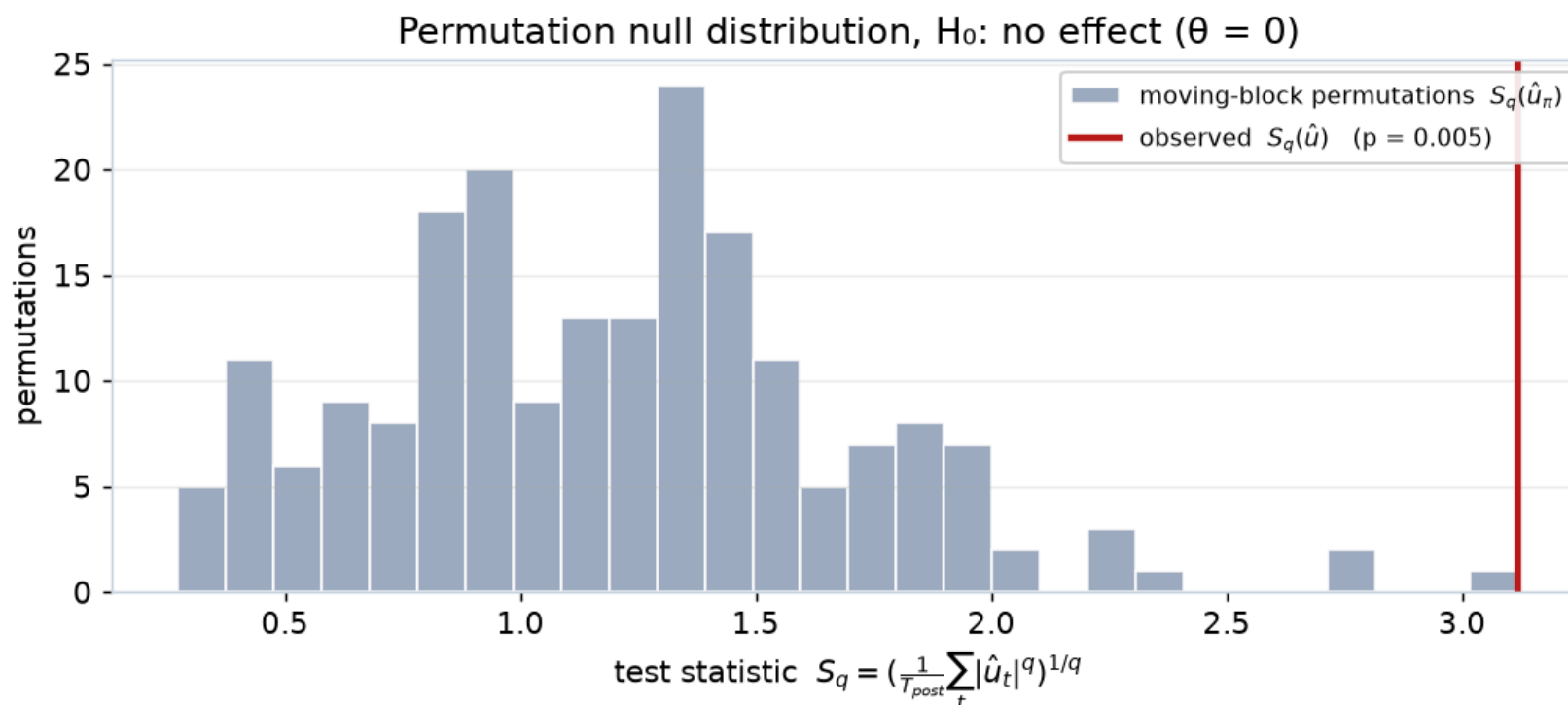
- Summarize the post-period residuals with a single number ($q = 1$ = mean absolute residual), so "the residuals look big" becomes something you can actually test:

$$S_q(\hat{u}) = \left(\frac{1}{T_{post}} \sum_{t > T_0} |\hat{u}_t|^q \right)^{1/q}$$

- Compare S_q to its distribution under **moving-block permutations** of the residual sequence — the serial-dependence-robust scheme from the last section.
- `panelib`'s `conformal_inf` implements exactly this, so you don't have to.
- Details: [*Appendix — Choosing the Test Statistic*](#)

Step 3 — the permutation p-value

$$p = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1}\{S_q(\hat{u}_\pi) \geq S_q(\hat{u})\}, \quad \text{reject } H_0 \text{ if } p \leq \alpha.$$

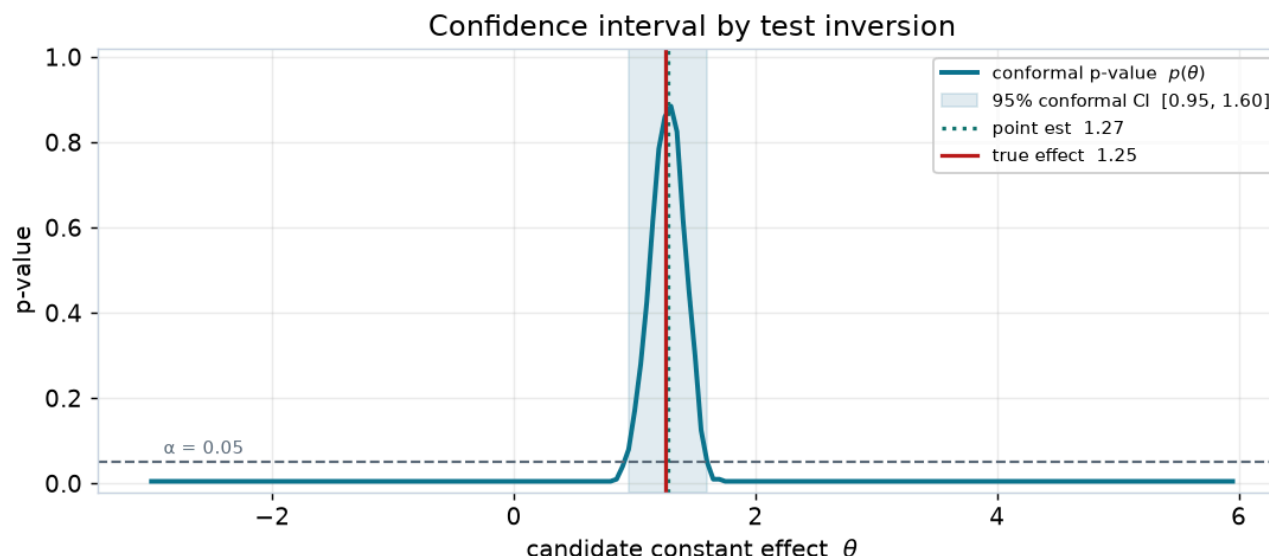


- **Careful:** this permutes residuals **across time** for one treated unit — not the Abadie placebo test, which permutes treatment **across units**.

Step 4 — confidence interval by test inversion

- Repeat the test over a grid of candidate effects θ and keep every value you **cannot reject** — that's a full confidence interval, not just a yes/no test:

$$CI_{1-\alpha} = \{ \theta : p(\theta) > \alpha \}$$



- The point estimate sits where $p(\theta)$ peaks; the CI is the set of effects consistent with "no break."

Why this matters

- **The payoff, one more time:** this works with **one treated unit and few pre-periods** — and is **exact** under exchangeability, **robust** to serial dependence and to a **misspecified** counterfactual (validity comes from residual symmetry, not a correct model).
- The fine print: under dependence the guarantee is approximate, and it needs the counterfactual's **error to be stable over time** — a model whose bias *trends* into the post-period breaks a symmetry no permutation can restore.
- Resolution matters too: with very few post-periods the permutation p-value is coarse — in the Panel deck's simulations (2 post-periods) coverage ran **80–85%**, not 95%.
- Contrast Abadie placebo inference: its p-value resolution is only $1/(J + 1)$ and it needs a large donor pool.
- This is the inference the **Panel Models** deck used to put DiD, SC, and SDID on the same footing.

Conclusion

- **Back to the big idea:** conformal inference buys an **honest, finite-sample guarantee** from **one assumption you can actually reason about** — exchangeability — and it degrades gracefully, rather than catastrophically, when that assumption is stressed.
- Reach for it to **wrap black-box predictors** you don't fully trust, and — the reason we built this whole deck — to do inference where classical SEs go silent: **a single treated unit over time**.
- The price: coverage is **marginal**, not conditional; and under dependence you must **block** or **weight** to keep the guarantee.
- Bottom line: start from the guarantee you can actually defend, then pick the score and the permutation that fit your data.

Appendix

Appendix: split vs. full conformal

- **Split (inductive)** conformal — the recipe here — spends part of the data on a calibration fold. Cheap (one fit), but the interval is random in that split and a bit wider.
- **Full** conformal refits for every candidate label — exact but expensive; **CV+ / jackknife+** (Barber et al. 2021) recover most of the efficiency without a held-out split.
- Same guarantee; the choice is **data efficiency vs. compute**.

Appendix: the coverage law

- Conditional on the calibration draw, the coverage of split conformal is a **random variable**:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C} \mid \text{calibration}) \sim \text{Beta}(k, n + 1 - k), \quad k = \lceil (n + 1)(1 - \alpha) \rceil.$$

- Its mean is $\geq 1 - \alpha$; the spread shrinks like $1/\sqrt{n}$ — the middle histogram made this concrete.
- The upper bound $1 - \alpha + \frac{1}{n+1}$ needs the scores to have **no ties** (continuous scores, or a random tie-break).

Appendix: Choosing a Conformity Score

- Any score $s(x, y)$ is valid for the guarantee — the only thing at stake in the choice is **efficiency** (set tightness), never validity.
- Common families beyond the absolute residual $|y - \hat{f}(x)|$:
 - **Normalized:** $s = |y - \hat{f}(x)| / \hat{\sigma}(x)$ — divides by a local scale so the score reflects *relative* surprise (the fix behind **conformalized quantile regression**, Romano, Patterson & Candès 2019).
 - **Quantile-based:** $s = \max\{\hat{q}_{\alpha/2}(x) - y, y - \hat{q}_{1-\alpha/2}(x)\}$ — built from two fitted quantile regressions.
 - **Classification margin:** $s = 1 - \hat{\pi}_y(x)$ — the standard score outside regression.
- Rule of thumb: pick the score that captures *where* your model is uncertain — a score blind to heteroskedasticity gives valid but needlessly wide sets exactly where the model is confident.

Appendix: Why Coverage Survives a Bad Model

- Split conformal's guarantee has two separable pieces: **validity** (coverage $\geq 1 - \alpha$) and **efficiency** (set tightness). The rank argument on "Why it works" only ever uses exchangeability — $\hat{f}$ never enters it.
- Extreme case: let $\hat{f}(x)$ be the training fold's unconditional mean, ignoring x entirely. The set is far wider than a well-fit model's — but it still covers $1 - \alpha$ of the time, since calibration recenters around whatever the score distribution actually is.
- This is why conformal is a popular *wrapper* around models you don't fully trust — validity doesn't require the wrapped model to be well specified, only that the calibration fold is exchangeable with the test point.
- Back to [*The Guarantee Holds Even When Your Model Is Wrong*](#)

Appendix: Conformal Sets as Test Inversion

- $p(y)$ is a genuine p-value: under exchangeability and $H_0 : Y_{n+1} = y$, $p(y)$ is (super-)uniform on $[0, 1]$ — the same guarantee classical hypothesis tests offer, with no distributional assumption beyond exchangeability.
- This makes conformal prediction a special case of **Neyman's construction**: build a valid test for every candidate value, then invert by collecting every value not rejected. Classical intervals from pivotal statistics are the parametric special case; conformal drops the pivot requirement entirely.
- The duality runs both ways: any exchangeability-respecting permutation test induces a valid conformal-style set by the same inversion — exactly the move the synthetic-control section makes. The permutation p-value there **is** this $p(y)$, with y replaced by a hypothesized treatment effect θ .
- Back to [*Conformal Prediction Is Hypothesis Testing in Disguise*](#)

Appendix: another route — prediction intervals

- Conformal is not the only distribution-aware option for synthetic control.
- **Cattaneo, Feng & Titiunik (2021)** build **prediction intervals** that separate two error sources: the in-sample **weight-construction** error and the out-of-sample **stochastic** error, with finite-sample guarantees.
- Rule of thumb: **conformal** tests a sharp null by permutation and inverts to a CI; **prediction intervals** model the two error sources directly. Different assumptions, complementary diagnostics.

Appendix: conformal for treatment effects, beyond panels

- **Individual treatment effects** — Lei & Candès (2021): conformal intervals for $Y(1) - Y(0)$ under unconfoundedness, using *weighted* conformal to bridge the treated/control covariate shift. The natural companion to the HTE deck.
- **Covariate shift** — Tibshirani, Barber, Candès, Ramdas (2019): weighted conformal stays valid when the test point comes from a different covariate distribution with a known (or estimated) likelihood ratio.
- **Drift in production** — Gibbs & Candès (2021): *adaptive* conformal updates the level α_t online to keep realized coverage on target under distribution shift — the monitoring use case for deployed models.

Appendix: The CWZ Framework — Assumptions

- Formally, CWZ needs the post-period residuals to be (approximately) **exchangeable with the pre-period residuals under H_0** — delivered by **stationarity** of the residual process plus **weak dependence** (mixing decays fast enough that a block permutation approximately preserves the joint law).
- The framework is agnostic to *how* the counterfactual is built: DiD, synthetic control (ADH), synthetic DiD, or a factor model all plug in as the same $\hat{Y}_{1t}(0)$ — only the residual's exchangeability under the null matters, not the estimator's asymptotics.
- **Validity vs. power** are cleanly separated: a poor counterfactual still yields a valid test (residuals stay symmetric under H_0), but a good one yields a *powerful* test — model quality shows up in power, never in size.
- *Continued next slide — comparisons and the multi-unit case.*

Appendix: The CWZ Framework — Comparisons

- **Abadie placebo** permutes treatment across *units* and needs a sizable donor pool for p-value resolution; **CWZ** permutes residuals across *time* for a single unit and needs enough pre-periods instead. Different resampling axis, same test-inversion logic.
- For **multiple treated units**, see the Panel Models deck's appendix on inference with many treated units — CWZ's single-unit permutation isn't the right tool once you have several.
- Back to *Causal Inference = Prediction + a Break Test*

Appendix: Choosing the Test Statistic

The summary statistic S_q trades off what kind of break it's sensitive to:

- $q = 1$ (mean absolute) — the deck's default; balanced power against a sustained, moderate shift.
- $q = 2$ (root-mean-square) — more sensitive to a single large spike in the post-period than to many small ones.
- $q = \infty$ (max absolute deviation) — detects a single sharp break, at the cost of power against a gradual trend.
- *Continued next slide — block length and the permutation count.*

Appendix: Choosing the Permutation Scheme

- **Block length** trades bias against variance: too short and the permutation shatters real local dependence (inflated size); too long and too few distinct block placements are left to permute (coarse, low-power p-values). Heuristic: set it near the residual series's estimated dependence range.
- **Number of permutations $|\mathbf{\Pi}|$** : exhaustive block permutations are cheap for short pre-periods; for longer series, Monte Carlo sampling from $\mathbf{\Pi}$ gives the same test with a controllable p-value resolution ($1/|\mathbf{\Pi}|$ sets the finest achievable p-value).
- This is exactly what `panelib`'s `conformal_inf` exposes as tunable arguments — the defaults above are its choices.
- *Back to [One Number, Tested Against Its Shuffled Twins](#)*

Literature Review of Related Papers

- **Conformal prediction — foundations**

- Vovk, Gammerman, Shafer (2005). *Algorithmic Learning in a Random World*. Springer — the founding monograph. [link](#)
- Shafer & Vovk (2008). "A Tutorial on Conformal Prediction." *JMLR* 9, 371–421. [link](#)
- Lei, G'Sell, Rinaldo, Tibshirani, Wasserman (2018). "Distribution-Free Predictive Inference for Regression." *JASA* 113(523). [link](#)
- Angelopoulos & Bates (2023). "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification." *Found. & Trends ML*. [link](#)

Literature Review (continued)

- **Adaptivity & beyond exchangeability**

- Romano, Patterson, Candès (2019). "Conformalized Quantile Regression." *NeurIPS*. [link](#)
- Barber, Candès, Ramdas, Tibshirani (2021). "Predictive Inference with the Jackknife+." *Ann. Statist.* 49(1). [link](#)
- Barber, Candès, Ramdas, Tibshirani (2023). "Conformal Prediction Beyond Exchangeability." *Ann. Statist.* 51(2), 816–845. [link](#)
- Tibshirani, Barber, Candès, Ramdas (2019). "Conformal Prediction Under Covariate Shift." *NeurIPS*. [link](#)
- Gibbs & Candès (2021). "Adaptive Conformal Inference Under Distribution Shift." *NeurIPS*. [link](#)

Literature Review (continued)

- **Conformal inference for causal panels / synthetic control**

- Chernozhukov, Wüthrich, Zhu (2018). "Conformal Inference Methods for Predictive ML with Dependent Data." *COLT (PMLR 75)*. [link](#)
- Chernozhukov, Wüthrich, Zhu (2021). "Conformal Inference for Counterfactual and Synthetic Controls." *JASA* 116(536). [link](#)
- Cattaneo, Feng, Titiunik (2021). "Prediction Intervals for Synthetic Control Methods." *JASA* 116(536). [link](#)
- Firpo & Possebom (2018). "Synthetic Control Method: Inference, Sensitivity Analysis and Confidence Sets." *J. Causal Inference* 6(2). [link](#)
- Lei & Candès (2021). "Conformal Inference of Counterfactuals and Individual Treatment Effects." *JRSS-B* 83(5). [link](#)

- **Companion code:** `panelib.py` (`conformal_inf`) in [statanomics](#).